

# **A Guided Search vision module for ACT-R: Implementation, tutorial, and validation against benchmark visual search data**

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## **Abstract**

Cognitive models that interact with visual displays depend on ACT-R's vision module to locate and encode objects. However, the stock vision module's core search mechanisms have remained largely unchanged for more than two decades and do not include visual acuity, feature-guided selection, eye position, or an adaptive decision to terminate an unsuccessful search. This paper's primary contribution is gs-vision, an extension of the ACT-R 7.31 vision module that integrates a Guided Search 6 priority map, Competitive Guided Search selection and identification, memory for rejected items, adaptive quitting, per-feature acuity, iconic memory, and EMMA eye movements. The tutorial explains the module's theoretical basis and provides a step-by-step guide to loading it, describing displays, issuing searches, collecting diagnostics, and fitting parameters from Python. We validate the module against reaction-time distributions and error rates from three benchmark visual-search tasks and compare it with the stock ACT-R vision module, PAAV, an ACT-R variant based on the posted Guided Search 6 engine, and seven trial-

level configurations from five model families under common scoring criteria. On test participants excluded from parameter fitting, the exploratory gs-vision refit obtained a mean cell quantile RMSE of 100.1 ms, compared with 122.5 ms for the next-best ACT-R vision-module account and 80.5 ms for the best trial-level model. This is the lowest numerical RT-quantile error among the ACT-R accounts tested, but validation selection favored the frozen gs-vision fit (test RMSE 127.7 ms). The refit followed inspection of the frozen test results, and high miss rates and long saccades remain limitations.

*Keywords:* ACT-R, visual search, Guided Search, cognitive architecture, eye movements, model validation, reaction time distributions

## **Introduction**

Cognitive architectures are computational theories that describe how perceptual, cognitive, and motor mechanisms work together to produce behavior (Anderson, 2007; Anderson et al., 2004; Newell, 1990). ACT-R is particularly useful for human-computer interaction because its models can inspect a display, move attention and the eyes, retrieve knowledge, and produce motor responses with explicit predictions about time and error (Byrne, 2001; Salvucci, 2001). These capabilities have enabled ACT-R models to explain behavior in menus, complex displays, and other interactive tasks. However, the quality of those predictions depends on how the architecture represents the search for task-relevant information.

The stock ACT-R vision module provides a reliable symbolic interface between a model and a display. A model can request an object with a specified color, value, or location, shift attention to the selected location, and encode the object. The module was not designed as a current theory of visual search. Its core search operations do not include several important features, including eccentricity-dependent acuity, salience-based competition, graded guidance

by target features, a representation of eye position in selection, or an adaptive rule for deciding that a target is absent. Modelers can create a production loop that locates, attends to, and tests one object at a time, but each iteration includes production and encoding costs that are substantially slower than human item-processing rates in many visual-search tasks (e.g., Fleetwood & Byrne, 2006; Wolfe et al., 2010).

Research on visual search has developed several mechanisms that can address these limitations. Guided Search describes selection through a priority map that combines bottom-up salience, top-down target guidance, selection history, value, and scene structure (Wolfe, 1994, 2021). Guided Search 6 further includes capacity-limited identification, memory for rejected items, functional visual fields, and adaptive quitting. Competitive Guided Search (CGS; Moran et al., 2013) provides a quantitative account of selection, identification, and quitting that reproduces benchmark RT distributions and error rates. Related fixation-based and active-vision accounts show that acuity and eye movements also shape search efficiency (Hulleman & Olivers, 2017; Kieras & Hornof, 2014; Kieras & Meyer, 1997).

Earlier work has brought some of these ideas into ACT-R. The Pre-Attentive And Attentive Vision module (PAAV; Nyamsuren & Taatgen, 2013) added eccentricity-dependent acuity, iconic memory, a Guided Search activation map, and a rule for terminating search. Other proposals added salience-based selection to ACT-R location requests (Byrne, 2006; Wiese et al., 2019). These contributions demonstrated that ACT-R's visual interface can be extended, but they preceded ACT-R 7 or did not evaluate full RT distributions and error rates across searches with different efficiencies. The stock ACT-R 7 vision module therefore still lacks an integrated, quantitatively validated account of current visual-search mechanisms.

This limitation matters beyond laboratory visual search. An ACT-R model that interacts with a menu, cockpit, air-traffic display, spreadsheet, or web page must find information before it can use that information. When search time is represented as a fixed encoding delay or an unconstrained symbolic request, the model cannot predict how target-distractor similarity, eccentricity, set size, eye position, or target prevalence changes performance. A more complete vision module can connect display design to predicted search behavior while preserving the cognitive and motor mechanisms already represented in ACT-R.

In this paper, we address this limitation by presenting *gs-vision*, an extension of the ACT-R 7.31 vision module. The module combines Guided Search 6 as its organizing framework, CGS as its quantitative selection and identification engine, EPIC- and PAAV-style acuity and iconic memory, and EMMA eye movements. It retains the stock buffers and requests and adds a full-search request that runs the item-level search process within the module. The included compatibility tests verify unchanged traces for the ACT-R tutorial models used in the test suite when *gs-vision* is enabled or disabled.

This tutorial addresses three research questions:

1. Can current visual-search mechanisms be integrated into ACT-R while preserving the stock vision interface for existing models?
2. Does *gs-vision* simulate benchmark visual-search RT distributions more accurately than existing ACT-R vision modules under a common evaluation protocol?
3. How can cognitive modelers use the module to represent search in laboratory and applied tasks, inspect its predictions, and fit it to behavioral data?

The paper makes three contributions. First, it provides a working ACT-R vision module that connects feature guidance, acuity, eye movements, identification, memory, and quitting in one

event-driven process. Second, it provides a practical workflow for loading the module, describing displays, issuing searches, logging model behavior, and fitting parameters. Third, it evaluates the module against held-out participants from three visual-search benchmarks and compares it with the stock ACT-R module, PAAV, an ACT-R Guided Search 6 variant, and seven trial-level configurations from five model families under the same objective. The evaluation also reports where the module fails, because those limitations define the appropriate scope of its current use. All reported analyses and tables can be regenerated from the public repository.

### Limitations of the stock ACT-R vision module

The stock vision module (ACT-R 7.31.4, vision.lisp version 11.1; Bothell, n.d.) maintains a *visicon*, a list of feature chunks that the experiment adds with `add-visicon-features`. A visual-location request specifies slot constraints such as `color red` or `:attended new`, optional ordinal constraints such as `screen-x lowest`, and a `:nearest` option. The module returns one matching chunk in 0 ms, breaking ties by recency of onset and then at random. A visual request with `move-attention` shifts attention to that location and, after `:visual-attention-latency` (85 ms), places an object chunk in the visual buffer. Four *finsts* mark recently attended objects for three seconds, so that `:attended nil` can exclude them. These are the stock module's search primitives. Table 1 summarizes their timing in a model that uses three productions to find, attend to, and test each candidate object.

**Table 1**

*Per-item timing of the stock vision module in a find, attend, test production loop*

| Component                    | Latency                    | Source                    |
|------------------------------|----------------------------|---------------------------|
| Production cycle             | 50 ms each, three per item | Default :dat              |
| visual-location request      | 0 ms                       | Vision module             |
| Attention shift and encoding | 85 ms                      | :visual-attention-latency |

| Component                           | Latency                      | Source                                           |
|-------------------------------------|------------------------------|--------------------------------------------------|
| Total per item                      | 235 ms for three productions | $3 \times 50 + 85$ ; 185 ms with two productions |
| Human, conjunction search           | 10-35 ms per item            | Wolfe et al. (2010)                              |
| Human, spatial-configuration search | 45-100 ms per item           | Wolfe et al. (2010)                              |

This production loop has three consequences. First, its 235 ms per-item cost exceeds the human conjunction and spatial-search slopes listed in Table 1. A direct request such as “color red” can avoid this cost, but it then returns a matching object without simulating the search process and produces target-absent responses in approximately 210 ms, about half the human value in the benchmark. Second, when several objects satisfy the request, the stock module does not rank them by their similarity to the target. Third, the module has no search-specific quitting rule. With four fixations and more than four objects, an absent trial requires an additional counting or memory strategy to terminate reliably. The model comparison below quantifies these limitations. EMMA (Salvucci, 2001), the eye-movement extension distributed with ACT-R, represents when and where the eyes move after a location has been chosen; it does not determine which location wins the search competition.

PAAV (Nyamsuren & Taatgen, 2013) replaced the location process with an abstract-location buffer served by iconic memory, eccentricity-dependent acuity, a Guided Search 4 activation map, and a pruning rule that lets a model report target absence without fixating every object. It reproduced conjunction slopes of approximately 20 ms per item for present trials and 54 to 73 ms per item for absent trials, as well as fixation counts in a comparative-search task. PAAV nevertheless selects one item per fixation, ties the item slope to the saccade rate, has no adaptive quitting mechanism for modeling prevalence and miss rates, and was developed for ACT-R 6. It was also not evaluated against full RT distributions and error rates. *gs-vision* builds

on PAAV’s acuity, iconic-memory layer, and integration pattern while replacing its selection, identification, and quitting processes. Because the original PAAV implementation does not load under ACT-R 7, the comparison uses a display-level mirror transcribed from its source code; we report this limitation with the corresponding results.

## The gs-vision module

## Theory

gs-vision implements Guided Search 6 (Wolfe, 2021) as the organizing architecture, Competitive Guided Search (Moran et al., 2013) as the arithmetic of selection, identification and quitting, EPIC-style acuity (Kieras & Meyer, 1997) with PAAV’s iconic memory for what is visible from where the eye is, and EMMA (Salvucci, 2001) for saccades. Figure 1 shows the pieces and the one-trial flow between them. Table 2 lists the equations.

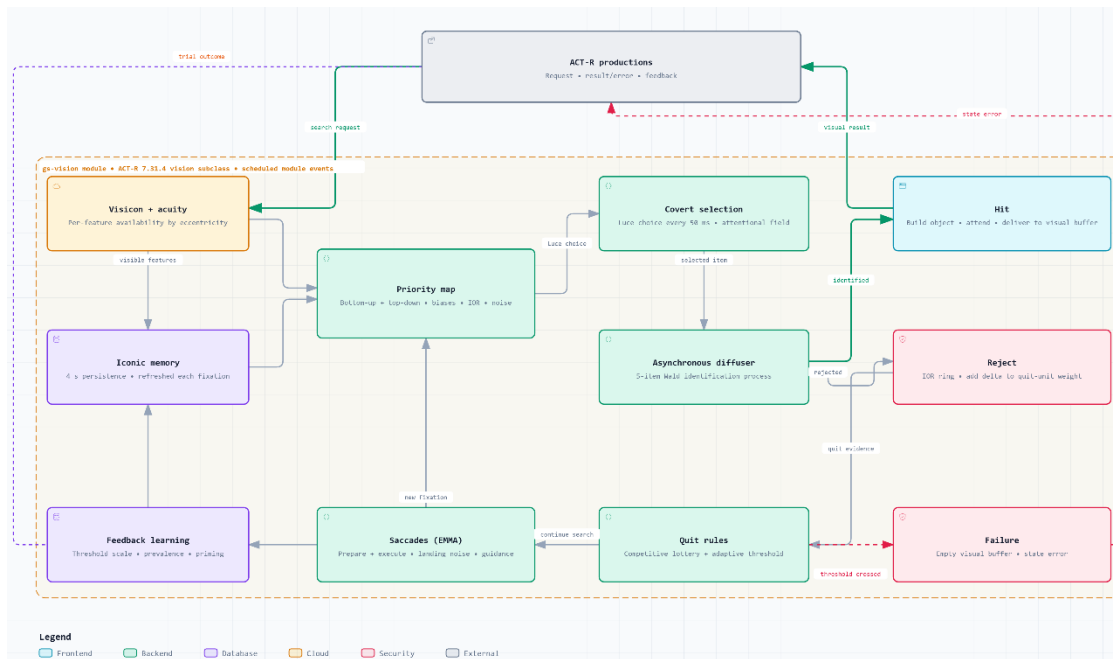


Figure 1. Architecture of gs-vision and the flow of one search. ACT-R productions issue a search request, receive a visual result or state error, and report trial feedback. The dashed orange boundary encloses the module’s scheduled processes. Solid green paths show the request and

successful result; the dashed red path returns state error. The diagram summarizes interacting processes, which can overlap in time.

*Priority map:* Every item in iconic memory receives a priority that is a weighted sum of bottom-up salience (local contrast in categorical feature channels, divided by inter-item distance, as in Guided Search 2), top-down guidance (categorical channel match to the guiding template; features that are not yet visible contribute an uncertainty term), a history term (priming traces per feature value decaying over about ten seconds), an optional value term and an optional scene prior, minus inhibition of return for recently rejected items, plus logistic noise. Guidance is restricted to *guiding features* (by default color, orientation, size, and luminance); every other feature, and shape above all, is compared only once an item is inside the identification stage. That one rule is what makes feature search efficient and spatial-configuration search inefficient with a single set of mechanisms.

*Covert selection and identification:* Every 50 ms by default, while a search is active and the identification stage has capacity (five items), the module selects one item by a probabilistic choice over the noisy priorities of the items inside the attentional functional visual field (8 degrees by default). Each selected item is identified after a Wald-distributed time (drift and threshold as in CGS), which completes when the item's required features are available at the current eye position; if they are not, the item waits for a saccade. A match ends the search with a hit; a mismatch rejects the item, adds it to the inhibition-of-return ring, and increments the quit unit.

*Quitting:* Two rules race. The competitive rule of CGS quits with probability equal to the quit weight divided by the sum of the quit weight and the guidance weights of every unresolved item, evaluated after every rejection. The adaptive rule of Guided Search 6 maintains a threshold

scale that multiplies the effective set size; when the number of rejections reaches it, new selection pauses, outstanding identifications drain, and the search quits. Feedback after each trial lowers the scale after a correct rejection and raises it after a miss, using the hybrid update in Table 2, and updates a running prevalence estimate. The hybrid subtracts a constant step after a correct rejection; the posted GS6 engine instead scales its correct-rejection and miss updates by prevalence and one minus prevalence, respectively.

*Eyes:* A saccade is triggered when an item outside the attentional field carries guidance that exceeds the best item inside it by a margin, or when nothing selectable remains inside it; when candidates remain within the attentional field, the destination is chosen from the exploratory field (12 degrees by default) by guidance minus a distance penalty. When that field is exhausted, the default fallback selects the strongest remaining guidance; :gs-explore-proximity enables a distance penalty for that fallback as well. Preparation, execution (20 ms plus 2 ms per degree), and landing noise follow EMMA. On landing, acuity is re-evaluated: each feature of each item is available with a probability given by the EPIC rule, item size compared with a threshold that grows linearly with eccentricity and has a per-feature slope, and available features are written into an iconic memory that persists for four seconds.

**Table 2**  
*Core equations of gs-vision*

| Mechanism                               | Equation                                                                                                                                                                       | Origin                            |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| Availability of feature $f$ of item $i$ | $P(\text{size}_i > N(\theta_f \cdot e_i, \sigma))$                                                                                                                             | EPIC (Kieras & Meyer, 1997), PAAV |
| Priority                                | $P_i = w_{BU}BU_i + w_{TD}TD_i + w_{HH}H_i + w_{VV}V_i + w_{SS}S_i - \text{IOR}_i + \varepsilon_i, \varepsilon \text{ logistic}$                                               | Guided Search 2 and 6             |
| Selection                               | $p(i) = \exp(\beta P_i) / \sum_j \exp(\beta P_j)$ , items inside the attentional field                                                                                         | CGS (stochastic)                  |
| Identification time                     | $\text{Wald}(\mu, \theta)$ ; mean $\theta/\mu$                                                                                                                                 | CGS                               |
| Competitive quit                        | $p(\text{quit}) = w_q / (w_q + \sum_{\text{unresolved}} w_j)$ ; $w_q$ increases by $\Delta$ per rejection, or $\Delta / \max(0.05, \text{scale})$ with :gs-adaptive-quit-delta | CGS                               |

| Mechanism        | Equation                                                                                                                                                                                               | Origin                                                                |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Adaptive quit    | $\text{rejections} \geq \text{scale} \times N_{\text{eff}}$ ; scale $\text{--}=\text{step}$ after a true negative, $\text{+}=\text{step} / (2 \cdot \text{goal} \cdot \text{prevalence})$ after a miss | GS6-inspired hybrid; posted engine uses additional prevalence factors |
| Saccade duration | 20 ms + 2 ms/deg, plus preparation and landing noise                                                                                                                                                   | EMMA                                                                  |

## Design decisions

Three design decisions determine how gs-vision is integrated with ACT-R: (1) the item-level search loop is done in the module, rather than productions; (2) gs-vision extends the stock vision module rather than replacing its behavior; and (3) each model run represents one simulated observer.

*The item-level loop runs inside the module:* Productions fire every 50 ms, which coincidentally matches Guided Search's selection rate but not its identification or fixation rates, and production overhead plus the 85 ms shift makes production-driven item search several times too slow. The declarative module set the precedent: retrieval is sub-symbolic and internal, and productions only request and receive. gs-vision does the same for search. A production sets the template; the module runs selection, identification, saccades, and quitting on its own scheduled events; the target object arrives in the visual buffer, or the buffer is left empty with state error, exactly like a retrieval failure.

*It is a subclass of ACT-R vision module:* The module class inherits from vision-module. The :vision module is undefined and redefined with the same buffer names and every stock parameter, which is rebuilt from the live parameter table rather than copied from vision.lisp. Ordinary requests are routed to the stock methods. This design preserves the connections to the motor module, AGI devices, the Environment, and EMMA. The included backward-compatibility test runs the ACT-R tutorial unit 2 and unit 3 models under the stock module and under gs-vision, both enabled and disabled, and requires identical traces line by line. This test

provides evidence for compatibility with the tested interface; it does not establish compatibility with every ACT-R model.

*Each model run is one simulated observer.* The adaptive threshold, priming traces, and prevalence window carry experience across trials. Between trials, use `gs-reset-search` to clear only trial-specific state; using `reset` clears this accumulated history, the experience across trials, and therefore represents a new observer.

## Implementation

The module contains approximately 2,000 lines of Common Lisp in five files. The entry point `load-gs-vision.lisp` loads ACT-R, then EMMA, then `gs-vision`, before any model is defined. A Python mirror (`reference/gs_hybrid.py`) implements the same mechanisms with independent random streams. We use this mirror for parameter fitting because Python allows for parallel processes, and we compare it with the Lisp module on identical displays after implementation changes. The validation section reports this Lisp-mirror agreement. A Python harness drives ACT-R through its remote interface, generates displays, records event times, fits parameters, and evaluates runs against human data.

## Tutorial: using `gs-vision` in a model

This section shows how to add `gs-vision` to a cognitive model and use it in an experiment. The example is the same model used for the ACT-R results reported below. To reproduce it, install ACT-R 7.31.4, Steel Bank Common Lisp, and the Python dependencies listed in `requirements.txt`. The repository does not distribute ACT-R itself; the setup instructions identify the required version and download location.

### Step 1: Load the module

The module must be installed before any model is defined, because ACT-R cannot undefine and replace the vision module once a model exists.

```
sbcl --load gs-vision/load-gs-vision.lisp
```

Inside a model, two parameters turn the pieces on. With `:gs-enabled nil` the new requests are refused and the module behaves as the stock one, which is useful for ablations.

```
(sgp :emma t :gs-enabled t)
```

### Step 2: Describe the display

Displays are added exactly as before with `add-visicon-features`. The module defines a chunk-type `gs-feature` that includes visual-location and adds the continuous slots hue (0-360), orient (degrees, 0 vertical), lum (0-1), and the categorical slot shape. Only the slots a modeler needs must be given: color is used when hue is absent, and size (in square degrees) is derived from width and height when it is absent. The listing below is what the harness sends for a red vertical bar, a green vertical bar and a digit.

```
(add-visicon-features
 '(isa (gs-feature) screen-x 512 screen-y 300 kind bar color red value bar
      shape bar hue 0 orient 0 lum 0.5 width 20 height 60 size 0.7)
 '(isa (gs-feature) screen-x 700 screen-y 420 kind bar color green value bar
      shape bar hue 120 orient 0 lum 0.5 width 20 height 60 size 0.7)
 '(isa (gs-feature) screen-x 300 screen-y 500 kind digit color white value two
      shape two hue 0 orient 0 lum 1.0 width 30 height 40 size 0.6))
```

Two slots are available for models with a pixel front end: salience replaces the module's internal bottom-up computation with an externally computed value, and prior supplies a scene prior. Both enter the priority map with their own weights.

### Step 3: Issue a search

The full search is one request. The slots name the template; the module decides which of them guide (by default color, orientation, size and luminance) and which are compared only after selection. Options restrict guidance to named slots or select the quitting rule.

```

+visual> isa gs-search color red orient steep ; conjunction search
+visual> isa gs-search shape two ; no guiding feature: spatial search
+visual> isa gs-search color red guide color ; restrict guidance to color
+visual> isa gs-search color red stop cgs ; adaptive | cgs | both (default)

```

Channel names such as steep, shallow, red and small are categorical, following Guided Search; a numeric value is mapped to the channel it falls in. The result arrives in the visual buffer as an ordinary visual object, or the search quits and the buffer is left empty with state error. A model therefore branches on two buffer tests, exactly as it does for retrievals:

```

(p respond-present
  =goal> isa trial state searching
  =visual>
  ?manual> state free
  ==>
  +manual> cmd press-key key "j"
  -visual>
  =goal> state responded)

(p respond-absent
  =goal> isa trial state searching
  ?visual> state error
  ?manual> state free
  ==>
  +manual> cmd press-key key "f"
  =goal> state responded)

```

For models that want to keep their own production loop, a guided location request exists as well. It ranks the visicon by the priority map instead of choosing at random and costs one selection interval rather than 0 ms; the usual filters (screen-x, :nearest, :attended) still apply.

```

+visual-location> isa visual-location color red orient steep :guided t

```

#### Step 4: Report the outcome

Adaptive quitting and priming need to know how the trial ended. The observer cannot know whether its response was correct, so the experiment tells it, and a production passes the outcome to the module after the keypress. This production is outside RT.

```

(p report-outcome
  =goal> isa trial state responded outcome =o
  ?visual> state free
  ==>
  +visual> isa gs-feedback outcome =o ; hit | miss | fa | tn
  =goal> state waiting outcome nil)

```

## Step 5: Read the diagnostics

Every selection, decision, saccade, fixation, quit and feedback is a trace event at :output medium, so it appears in the standard trace and in the Environment:

```
GS-SEARCH start template (COLOR RED ORIENT STEEP) n-eff 6. qt 1.00
GS-SELECT VISICON-ID2 priority 1.415
GS-DECIDE VISICON-ID2 reject
GS-SACCADE 512. 384. -> 700. 500. amp 11.0
GS-FIXATE 706. 495.
GS-DECIDE VISICON-ID9 hit
GS-FEEDBACK HIT qt 1.000 prevalence 0.50
```

Commands expose the same information to a harness: `gs-search-stats` returns fixations, rejections, elapsed time and the quit reason for the last search; `gs-fixation-log` returns stationary intervals as (time x y duration); `gs-event-log` returns every timestamped event; `gs-state` returns the threshold scale, the prevalence estimate and the feedback count. `gs-reset-search` clears per-trial state between trials while preserving learning, and `gs-benchmark-gaze` places the eye at a fixation cross without charging time, as an experiment's fixation cross does.

## Step 6: Drive the experiment from Python

The repository's harness uses ACT-R's own remote interface (the `actr.py` client shipped with the tutorial). One trial is: reset the search, place the gaze, add the display, arm the goal, run until the keypress, then send feedback. The listing below is a schematic of `harness/run_batch.py`; the full harness also installs the keyboard device and keypress monitor.

```
actr.call_command("gs-reset-search")
actr.delete_all_visicon_features()
actr.call_command("gs-benchmark-gaze", cx, cy)
actr.add_visicon_features(*display.visicon_features())
actr.call_command("gs-trial-setup", task, color, orient, shape)
actr.run(10) # monitor records keypress; loop returns later
outcome = score(display, key_pressed)
actr.call_command("gs-trial-feedback", outcome)
actr.run(2) # feedback production and intertrial interval
stats = actr.call_command("gs-search-stats")
fixations = actr.call_command("gs-fixation-log")
```

`gs-trial-setup` and `gs-trial-feedback` are two Lisp functions defined at the bottom of the model file and registered as remote commands; they write the trial's template and outcome into

the goal chunk. RT is measured from stimulus onset to the first valid keypress, recorded by a keypress monitor; the return of the event loop is about 90 ms later and is never used as RT.

### Step 7: Fit parameters

The module's search parameters are set with `sgp` and mirrored one-to-one in the Python `GSParams` record. The harness validates a complete configuration, sends it to ACT-R, reads every value back, and hashes the accepted values into the run record, so that a run can always be traced to the parameters it actually used. Fitting uses differential evolution (Storn & Price, 1997) on the mirror with the objective of the Validation section, and the selected configuration is then run in ACT-R at fresh seeds. The Open Practices section gives commands for regenerating reports from archived runs and points to the fitting entry points.

Table 3 lists the parameters a modeler is most likely to touch. The complete list, with the stock vision parameters that continue to work, is in the module's README.

**Table 3**  
*Principal gs-vision parameters*

| Parameter                                    | Default                                             | Role                                                                  |
|----------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------|
| :gs-guiding-features                         | (color orient size lum)                             | Features that may guide selection; all others are identification-only |
| :gs-select-interval                          | 0.050 s                                             | Time between covert selections                                        |
| :gs-diffuser-capacity                        | 5                                                   | Items identified at once                                              |
| :gs-choice-beta                              | 4.0                                                 | Inverse temperature for Luce choice over priorities                   |
| :gs-id-drift, :gs-id-threshold, :gs-id-sigma | 0.25, 0.03, 0.1                                     | Wald identification: mean $\theta/\mu$ , noise $\sigma$               |
| :gs-id-error                                 | 0.0                                                 | Probability that an identification decision flips                     |
| :gs-quit-delta                               | 0.02                                                | Competitive quit weight per rejection                                 |
| :gs-qt-init, :gs-qt-step, :gs-error-goal     | 1.0, 0.05, 0.08                                     | Adaptive quitting                                                     |
| :gs-memory                                   | 4                                                   | Inhibition-of-return ring size                                        |
| :gs-attn-fvf, :gs-explore-fvf                | 8, 12 deg                                           | Attentional and exploratory fields                                    |
| :gs-saccade-margin, :gs-saccade-trigger      | 0.25 priority units; 0 deg (disabled)               | Guidance margin; optional nearest-item distance trigger               |
| :gs-acuity-theta                             | color .10, orient .20, shape .40, size .20, lum .10 | Acuity slope per feature                                              |

| Parameter                                     | Default               | Role                                                                                   |
|-----------------------------------------------|-----------------------|----------------------------------------------------------------------------------------|
| :gs-w-bu, :gs-w-td, :gs-w-h, :gs-w-v, :gs-w-s | 0.5, 1.0, 0.3, 0, 1.0 | Priority-map weights                                                                   |
| :gs-noise                                     | 0.2                   | Logistic noise on priorities                                                           |
| :gs-priming-tau                               | 10 s                  | Decay of feature priming                                                               |
| :gs-iconic-span                               | 4 s                   | Persistence of a feature in iconic memory                                              |
| :gs-value-hook                                | nil                   | Function or remote command supplying a value term per location                         |
| :gs-onset-latency                             | 0.0 s                 | Additional delay before the first selection, beyond :gs-select-interval                |
| :gs-adaptive-quit-delta                       | nil                   | Divide the competitive quit increment by $\max(0.05, \text{adaptive threshold scale})$ |
| :gs-explore-proximity                         | nil                   | Apply a distance penalty to destination choice when the attentional field is exhausted |

### Validation against benchmark human data

We assess how well each model reproduces response-time distributions and accuracy in the human visual-search benchmark. All comparisons use the same participant splits, RT filtering, and scoring criteria, with parameters fitted to training participants. The primary gs-vision selection was frozen using validation participants before test evaluation. The subsequent refit and model comparison followed inspection of those test results and are exploratory. Simulation details and optimization budgets differ across model classes, as specified below; trial-level baselines do not simulate displays or eye movements.

### Data

Wolfe et al. (2010) published trial-level RTs and accuracy for three tasks that span the range of search efficiency: a feature search (red vertical target among green vertical bars), a color-by-orientation conjunction search (red vertical among red horizontal and green vertical), and a spatial-configuration search (a digit 2 among 5s), each at set sizes 3, 6, 12 and 18, target present on half the trials, with 9, 10 and 9 observers who each completed about 4,000 trials. The archive contains 111,777 trials, and our harness’s importer retains every one of them, records the

SHA-256 of each source file, and fails on any malformed line. Following the source paper's methods, RT analyses use correct trials between 200 and 4,000 ms (feature, conjunction) or 200 and 8,000 ms (spatial), and accuracy uses every validated trial. Participants are weighted equally; a cell's human quantile is the mean over participants of that participant's quantile.

## Protocol

Participants were split within each task into training, validation and test groups of 5/2/2 (feature), 6/2/2 (conjunction) and 5/2/2 (spatial) with a fixed seed, before any model was fit. Parameters were fitted on the training participants, candidates were selected on the validation participants with fresh seeds, the selection was frozen, and the frozen model was then run in ACT-R with two further seeds at 1,000 retained trials per cell (2,000 per cell in total, 24 cells per split) and scored once against the test participants. The fitting objective was the mean over the 24 cells of the RMSE between model and human RT quantiles at .1, .3, .5, .7 and .9, plus 1,000 ms times the absolute difference in error rate, plus 2,000 ms times the timeout rate in each cell. These terms are averaged over cells; reported quantile RMSE excludes the error and timeout penalties. Simulated observers ran the experiment as the human observers did: 30 synthetic practice trials before each block of 300 retained trials, with a final block of up to 400, accuracy feedback after each keypress, a two-second interval before the next display, and a fixation cross at which gaze was placed before each trial. Because the source files do not contain item coordinates, displays reproduce the published geometry with random placement.

The report of every run states three summary numbers: mean cell quantile RMSE (ms), the number of the six RT-by-set-size slopes (three tasks by present and absent) that fall within 5 ms per item of the human slope, and the largest miss-rate error in percentage points. Targets set before the work were 40 ms, 6 of 6, and 3 points. The Lisp module and the Python mirror were

compared on identical displays in every cell with a Bonferroni-corrected z-test on search time; the frozen run has 0 of 24 exceedances with a maximum  $|z|$  of 1.47.

### Results of the frozen fit

The primary fit shares all settings across the three tasks; the module therefore has to derive the differences between tasks from the displays and the templates alone. Six parameters were fit (identification threshold, inhibition-of-return memory, selection interval, bottom-up weight, attentional field, competitive quit increment) with two small differential-evolution searches (four generations and an 18-member population per restart). Table 4 gives the frozen results on all three splits, together with the module at its defaults and with the exploratory refit described next.

**Table 4**

*gs-vision runs in ACT-R against the human splits (2,000 retained trials per cell, no timeouts)*

| Run                 | Fit parameters | Split       | Mean RT RMSE (ms) | Quantile RMSE (ms) | Slopes within 5 ms/item | Largest miss error (points) |
|---------------------|----------------|-------------|-------------------|--------------------|-------------------------|-----------------------------|
| Module defaults     | 0              | train       | 465.9             | 351.9              | 3/6                     | 11.5                        |
| Module defaults     | 0              | validation  | 593.4             | 411.1              | 5/6                     | 15.6                        |
| Module defaults     | 0              | test        | 529.1             | 365.4              | 3/6                     | 12.4                        |
| Frozen shared fit   | 6              | train       | 160.7             | 151.5              | 3/6                     | 10.1                        |
| Frozen shared fit   | 6              | validation  | 123.7             | 143.1              | 5/6                     | 16.5                        |
| Frozen shared fit   | 6              | <b>test</b> | <b>108.0</b>      | <b>127.7</b>       | <b>2/6</b>              | <b>10.9</b>                 |
| Refit (exploratory) | 13             | train       | 60.0              | 65.7               | 5/6                     | 13.2                        |
| Refit (exploratory) | 13             | validation  | 191.6             | 173.3              | 2/6                     | 14.7                        |
| Refit (exploratory) | 13             | test        | 102.5             | 100.1              | 4/6                     | 11.3                        |

The frozen model reproduces the qualitative structure of the benchmark: flat feature-search functions, a conjunction slope near the human value on absent trials, and spatial-configuration slopes that rise with set size on both trial types, with absent RTs longer than present RTs and a spatial absent slope more than twice the present slope. Its quantitative misses on the test participants are a feature search that is 54 to 118 ms too fast across cells, a

conjunction absent function that is too steep (39 versus 33 ms per item) and starts too late, spatial-configuration slopes that are too shallow (78 and 32 versus 98 and 45 ms per item), and miss rates that rise too steeply with set size in conjunction and spatial search (10.8 and 20.9 percent at set size 18 against 6.3 and 10.0). It produces no false alarms at all, because identification in this version is a timer rather than a decision.

### **An exploratory refit**

The frozen run's own trial and fixation logs identified six causes for those misses: an excessively fast leading edge in the RT distribution; an identification noise that was a fixed constant, forcing a coefficient of variation of 1.3 on every identification; a mandatory saccade at small set sizes in spatial search, because shape was resolvable only within about four degrees; an adaptive quit controller that, in spatial search, had drifted to fifty times its starting scale and never bound, so that every miss came from the competitive rule; the structural impossibility of false alarms; and saccades that exceeded the attentional field because the covert loop exhausted the field before the eye moved. Six parameters were added to address them (Table 3: :gs-id-sigma, :gs-id-error, :gs-onset-latency, :gs-adaptive-quit-delta, :gs-explore-proximity, :gs-saccade-trigger), all defaulting to the frozen behavior so that the frozen results are unchanged, and thirteen parameters were refitted with a larger search (2,418 evaluations per completed restart). Two policy configurations were examined: both enabled :gs-adaptive-quit-delta and :gs-explore-proximity, and the trigger configuration additionally fixed :gs-saccade-trigger at 6 degrees. These policy settings were not among the thirteen continuously fitted parameters. Because the frozen test results had been inspected before this refit, it is exploratory, and Table 4 labels it so.

The refit improves the test fit from 127.7 to 100.1 ms, passes four of six slopes, and meets the 40 ms quantile target in seven of eight feature-search cells; conjunction-absent quantile RMSE improves about two- to four-fold at set sizes 12 and 18, with smaller improvements at set sizes 3 and 6 (Figure 2). Its largest miss-rate error increases from 10.9 to 11.3 points, and on validation participants it is *worse* than the frozen fit (173.3 versus 143.1 ms). Validation selection favored retaining the frozen shared fit; the trigger refit is reported as an exploratory candidate, not the validation-selected replacement. Differences between participant groups provide one explanation: the two validation participants in spatial search are 20 percent faster than the training group, and the two in feature search 25 percent faster, while the two in conjunction search are slower. A model fitted to the training participants' speed loses on validation; the frozen fit had undershot the training participants by about the same amount and consequently scored better on validation. With two participants per validation or test split, selection is sensitive to the sampled participants; the archived evaluation therefore reports both refit candidates on all three splits.

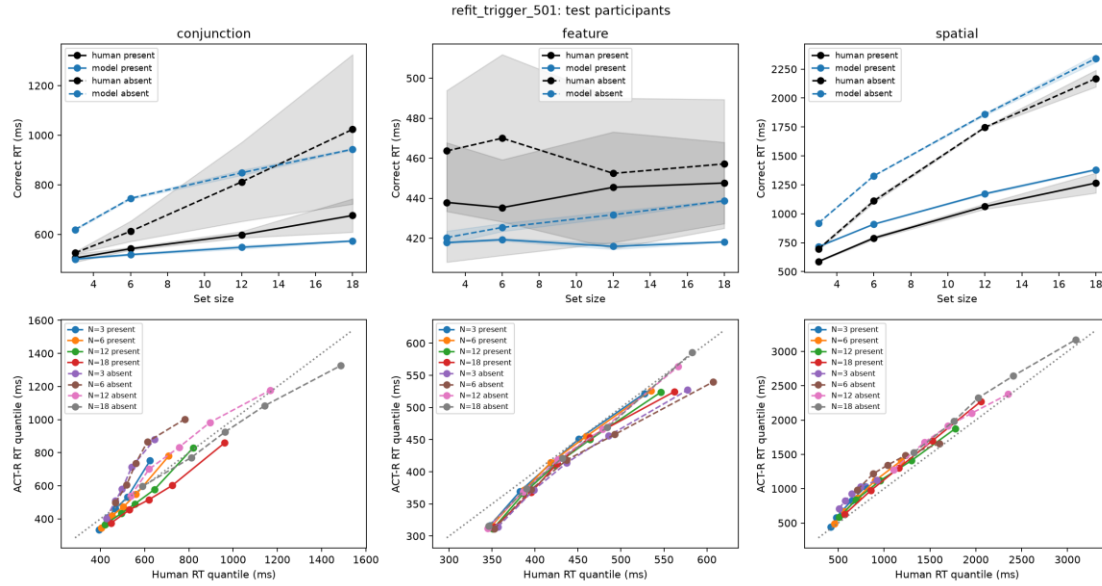


Figure 2. The exploratory refit of gs-vision run in ACT-R against the test participants. Top: mean correct RT by set size (shaded bands are the range across participants). Bottom: model against human RT quantiles (.1 to .9) for each of the eight cells per task; the dotted line is identity.

### Mechanism evidence: ablations

Fixed-parameter ablations around the module defaults (seeds 321 and 322; 300 retained trials per set-size-by-presence cell across seeds) examine the eye and quitting policies (Table 5). Restoring a second recognition delay adds 43 to 49 ms to mean present-trial RT across tasks; it leaves absent RT unchanged. Reverting the saccade policy raises the spatial fixation count from 6.1 to 12.9 per trial and spatial absent RT from 2.5 to 4.4 s, and produces timeouts in this ablation set. Raising the bottom-up weight from 0.5 to 3.0 raises conjunction absent RT from 1.1 to 2.8 s and the conjunction fixation count from 2.5 to 6.4. These default-parameter ablations differ from the frozen fitted configuration used for the secondary ACT-R manipulations below.

**Table 5**

*Ablations around module defaults (mirror; absent-trial RTs and spatial fixation counts across both target conditions)*

| <b>Configuration</b>               | <b>Conjunction<br/>absent RT (ms)</b> | <b>Spatial absent<br/>RT (ms)</b> | <b>Spatial fixations<br/>per trial</b> |
|------------------------------------|---------------------------------------|-----------------------------------|----------------------------------------|
| Module defaults (revised policies) | 1,146                                 | 2,510                             | 6.1                                    |
| Extra recognition delay restored   | 1,146                                 | 2,510                             | 6.1                                    |
| Old saccade policy                 | 1,707                                 | 4,417                             | 12.9                                   |
| Old quit weights                   | 1,192                                 | 3,218                             | 7.2                                    |
| Bottom-up weight 3.0               | 2,812                                 | 2,510                             | 6.1                                    |
| Threshold step 0.005               | 973                                   | 2,116                             | 5.4                                    |

## Eye movements

The benchmark has no eye data, so the module’s fixations were compared with the human values collected by Wu and Wolfe (2022) in a T-among-L foraging task whose data are on the Open Science Framework, with the caveat that foraging episodes and search-request-to-visual-result windows are not the same measurement. The human participant averages are 16.0 fixations per episode, 249 ms per fixation, 5.4 degree saccades, and a 9.5 percent refixation rate. The frozen gs-vision run produces stationary durations of 125 ms (feature), 186 ms (conjunction) and 142 ms (spatial), 4.6 fixations per spatial trial, and saccade amplitudes of 11.6 to 13.6 degrees; the refit lengthens the durations to about 200 ms in feature and conjunction search with mean amplitudes of 9.7 degrees in feature search, 12.1 in conjunction search, and 12.3 in spatial search. The refit’s spatial fixation duration remains 149 ms. Relative to the unmatched foraging reference, fixation durations are shorter and saccades are approximately twice as large; task differences prevent a direct accuracy judgment. A mechanism implicated by the model logs is that the covert loop clears the attentional field before the eye moves, so the next saccade goes to the best remaining item wherever it is. A saccade trigger that lets the eye start moving while covert selection continues was one of the refit’s additions; the reported trigger configuration

fixed it at 6 degrees, which did not resolve the amplitude discrepancy. Scanpath similarity to the human data (ScanMatch, MultiMatch) could not be computed for a matched experiment, because the foraging data lack the stimulus rotations and onset timestamps that a matched simulation needs; the human split-half consistency values are reported in the repository as descriptive consistency estimates, not a ceiling for the unmatched benchmark.

## Secondary phenomena

Three secondary effects were tested using the frozen shared fit, with a separate bottom-up-weight manipulation (Table 6). *Attentional capture* by an irrelevant salient singleton costs 7.6 ms in ACT-R at the frozen fitted weight of 1.049 and 23.0 ms at a weight of 3.0, against 39.4 ms in the human reference (Adam et al., 2021). The model uses an orientation target where the human study used a shape target, so this is calibration rather than a matched replication. *Priming of pop-out* is not established: the known-template switch-minus-repeat contrast is  $-2.9$  ms. In the unknown-color protocol, it is  $-6.2$  ms with history and  $-22.5$  ms without history, a change of  $-16.3$  ms that strengthens *the reversed* effect. At 10% rather than 50% target prevalence, model misses rise by only 1.5 points and absent responses *slow* by 87 ms. This fails to reproduce the combination of higher misses and faster absent responses reported by Wolfe et al. (2005). The hybrid's feedback rule differs from the posted GS6 engine: its correct-rejection step is constant, whereas the posted engine multiplies that step by prevalence and its miss update by one minus prevalence. The posted engine also fails the intended prevalence pattern in the separate replication. These observations do not establish that the two implementations fail for the same reason.

**Table 6**

*Secondary manipulations using the frozen shared fit, with a bottom-up-weight variant (ACT-R; 95% bootstrap intervals over two simulation seeds)*

| Effect                                                      | Model                  | Human reference                                                             |
|-------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------|
| Singleton capture cost, bottom-up weight 1.049 (frozen fit) | 7.6 ms [2.5, 12.7]     | 39.4 ms [31.6, 48.0] (Adam et al., 2021)                                    |
| Singleton capture cost, bottom-up weight 3.0                | 23.0 ms [21.7, 24.2]   | same                                                                        |
| Known-template priming, switch minus repeat                 | -2.9 ms [-10.0, 4.2]   | 20-60 ms (Maljkovic & Nakayama, 1994)                                       |
| Prevalence 10% vs 50%, change in misses                     | +1.5 points [1.3, 1.7] | Higher misses at low prevalence (Wolfe et al., 2005; qualitative reference) |
| Prevalence 10% vs 50%, change in absent RT                  | +87.3 ms [77.1, 97.5]  | faster                                                                      |

### Comparison with other models

A prospective researcher needs to know whether the added mechanisms improve accuracy relative to the available alternatives. We therefore scored 18 model configurations from nine model families, plus a human-reference baseline using the same human participants, splits, trimming rules, objective, and evaluation code (Tables 7 and 8; Figure 3). The comparison was conducted after the frozen test results had been inspected and is therefore exploratory. The ACT-R rows include direct runs of gs-vision and gs6-vision, a timing mirror of the stock module, and a display-level mirror of PAAV. Seven trial-level configurations represent five published model families. The fitted baseline and mirror configurations were optimized on training participants with differential evolution (Storn & Price, 1997; up to 60 generations, a population of 8 per dimension, and 200 trials per cell), then evaluated using two fresh seeds with 1,000 trials per cell per seed. The baseline driver retains the training-objective winner; it does not perform a separate validation-based candidate selection. Existing ACT-R runs retain their original fitting budgets and selection histories.

## **gs-vision**

gs-vision is the main model developed in this paper. It is an ACT-R vision module that operates on item locations and features, combines bottom-up salience and top-down guidance in a priority map, selects and identifies several items covertly, remembers rejected items, and races competitive and adaptive rules for ending an unsuccessful search. Per-feature acuity and iconic memory determine what information is available at each eye position, and EMMA supplies saccade timing and landing variability. The model returns a visual object to the ACT-R production system, which then makes the response through ACT-R's motor module. Tables 7 and 8 show three configurations of this same architecture: the unchanged module defaults, the prospectively frozen six-parameter fit, and the exploratory 13-parameter refit.

## **Stock ACT-R vision module**

The stock ACT-R module was driven by the ordinary find, attend, test loop with the same measured response stage. With its defaults (four finsts, 85 ms attention shift, three productions per item, a counting strategy so that absent trials terminate) its conjunction search runs at 47 to 57 ms per item present and 118 ms per item absent (human 11 and 33), its spatial search at 94 to 119 and 235 ms per item (human 45 and 98), and its feature search answers "absent" in 210 ms; its quantile RMSE on the test participants is 501 ms (Figure 4). Letting its attention latency, production count, finst count and response times vary brings it to 137 ms, and does so by driving the attention latency to 6 ms and the finst count to 19, which is no longer the stock module. These rows are a mirror of the module's documented timing rather than a run in ACT-R, and the mirror reproduces that timing exactly in a unit test.

## PAAV

PAAV was scored the same way, because it runs only on ACT-R 6 and porting its 7,000 lines to the current visicon, device and module interfaces was out of scope. Its mirror replicates the mechanisms of PAAV (paav-visual-module, version 0.98e):

Per-feature acuity (a feature of an item of size  $s$  at eccentricity  $e$  is visible when  $s > aje^2 - bje$ , with the posted  $a$  and  $b$  per feature and the noise term disabled as in the source), iconic memory with four-second persistence, bottom-up activation as binary feature dissimilarity over one plus the square root of the pixel distance, top-down activation of 1, 0.5 or 0 per template feature for a match, an invisible feature, or a mismatch, selection of the highest  $(1.1) \cdot BU + (0.45) \cdot TD + noise$  among unattended items, permanent attended marks, the visual decision threshold that prunes candidates no farther from gaze than the last attended object whose top-down activation does not exceed it and declares absence when nothing remains, saccades of 20 ms plus 2 ms per degree with landing noise, 50 ms encoding, and three 50 ms productions per item. It runs on the same displays as gs-vision, so acuity and distance act on it as they do on the module. With the posted values PAAV scores 383 ms on the test participants. Feature search fits (71 ms; the threshold rule quits after one fixation on absent trials), but conjunction absent slopes are three times the human value (94 versus 33 ms per item), spatial-configuration search is nearly efficient (5 and 23 versus 45 and 98 ms per item), because shape is resolvable within about nine degrees and the binary contrast rule makes the 2 pop out among 5s, and it misses 10 percent of targets in feature and spatial search, because items in the corners of a 22.5 degree field have no visible feature from the fixation cross and are not in iconic memory when the search quits. Fitting nine parameters (noise, the two map weights, encoding time, production count, an acuity scale and the response stage) brings it to 275 ms; the fit does so by removing bottom-up

activation (weight 0.03) and raising noise, which repairs conjunction search (28 ms per item absent) and the miss rates but leaves spatial search at 6 and 14 ms per item, and 3.6 percent misses at set size 18 against 10. A one-item-per-fixation architecture with a pruning rule cannot produce an inefficient search on a display where the target's shape is visible from a few fixations away, which is the limitation the Guided Search architecture removes.

### **gs6-vision**

A Guided Search 6 variant replaces the CGS engine with Wolfe's posted Guided Search 6 simulation: a two-bound diffusion per item stepped every 10 ms, an adaptive start point, the quit-signal diffuser with the MATLAB's feedback rules, diffuser-only memory, and Guided Search 2's dual orientation channels. It was built to answer "how close the module can be made to Wolfe's own numbers". The engine does not fit (795 ms untuned; 313 ms with only the spatial layer fitted around it). With its rates fitted, it reaches 122.5 ms on the test participants, about the frozen gs-vision fit, but with a drift, quit increment and selection interval far from the posted values. It produces false alarms at a plausible rate (1.3 percent against 2.3), closer than the gs-vision refit's 0.29 percent, and misses approximately 18 to 19 percent of targets at set size 3 in conjunction and spatial search, because its quit threshold scales with set size (Figure 5). Its absent-to-present slope ratio in conjunction search is 7, where the human value is 3.

The remaining comparison models are trial-level simulations. Each receives the task, set size, and target presence and returns a response and RT. They have no display, eye, or acuity function. Hence, they can be compared only on RT quantiles and errors. The following sections describe the operational implementations used in this comparison.

### **Serial self-terminating search (FIT)**

The serial self-terminating model is an operational baseline derived from Feature Integration Theory (Treisman & Gelade, 1980). Feature search uses one preattentive detection stage regardless of set size. Conjunction and spatial-configuration search inspect items sequentially in random order without replacement. A present trial ends when the target is found, while an absent trial exhausts the display. This produces the model's characteristic prediction that the absent slope is about twice the present slope. Identification misses and response flips provide its two error sources. This is a trial-level model. Meaning, it only receives the task, set size, and target presence, and does not simulate a display or eye movements.

### **Competitive Guided Search (CGS)**

Competitive Guided Search (Moran et al., 2013) treats items and a quit unit as competitors in a weighted race. Every distractor has unit weight, the target has a task-dependent guidance weight, and selection is proportional to these weights. Selecting an item incurs a noisy Wald identification time. Rejecting a distractor removes it from the race and increases the quit unit's weight, so the probability of stopping grows as the search continues. Tables 7 and 8 report two fitting protocols for the same mechanism. In "CGS, shared timing," identification and response timing are shared across tasks while target guidance and quitting vary by task. In "CGS, per task," all eight parameters are fitted separately for feature, conjunction, and spatial search, matching the more flexible protocol used in the original CGS analysis. Like the other trial-level models, CGS has no stimulus geometry, acuity, or eye movements.

### **Fixation-based search (H&O)**

The fixation-based baseline represents the model by Hulleman and Olivers (2017), in which the useful unit is a fixation rather than an individual item. Each fixation takes a variable

amount of time and samples up to a task-specific number of items within a functional visual field. The model remembers the items sampled during a fitted number of recent fixations. After that window, they can be sampled again. It responds “present” when the target is detected and “absent” after the accumulated samples cover a fitted proportion of the display. Early stopping and imperfect detection produce misses. Because this implementation has no item coordinates, its fixations represent sampling episodes and not the simulated gaze positions.

### **Parallel race with a capacity exponent**

The parallel-race baseline represents the parallel alternative described by Townsend and Ashby (1983) and compared with CGS by Moran et al. (2016). All items begin processing simultaneously, and each has a noisy Wald finishing time. A capacity exponent slows every item’s processing rate as set size grows, while a task-specific guidance factor gives a present target an advantage over distractors. A present response occurs when the target finishes, and an absent response waits for the slowest distractor. The model therefore explains set-size effects through divided processing capacity rather than serial selection. A target-miss probability and a response-flip probability generate errors.

### **Guided Search 6 trial-level engine**

The Guided Search 6 trial-level model replicates Wolfe’s posted simulation (Wolfe, 2021) using Python instead of ACT-R, and does not take display, acuity, and eye-movement layers used by gs6-vision into account. Items enter a capacity-limited diffuser and accumulate noisy evidence toward target or distractor bounds, while a separate quit signal races toward its threshold. Feedback modifies the identification start point after hits and false alarms and adjusts the quit threshold after misses and true negatives. Since there is no priority map, a fitted task-specific target weight represents guidance. “GS6 engine, as posted” fixes the diffusion and

quitting rates at Wolfe’s values and fits only guidance and response timing; “GS6 engine, rates fitted” also fits those engine rates. Comparing these rows with gs6-vision illustrates performance under different spatial and architectural constraints. The separately fitted rows do not isolate a single embedding cost.

### Human-reference baseline

The “training participants’ own averages” row is not a cognitive model and has no fitted parameters. It treats the mean cell quantiles and error rates of the training participants as predictions for the held-out test participants. Its score shows how much the participant groups differ under the same metric and gives a useful scale for interpreting small differences between fitted models. It is a reference for cross-participant variability, not an estimate of a theoretical lower bound.

### Comparison results

The remaining rows of Tables 7 and 8 are parameter configurations of the models described above, not additional architectures. The tables report their held-out reaction-time, slope, and error fits; Figures 3 through 5 summarize the principal differences.

**Table 7**

*Reaction-time fit for all models on the held-out test participants (quantile RMSE in ms)*

| Model                                      | ACT-R status           | Fit parameters | Overall      | Feature | Conjunction | Spatial |
|--------------------------------------------|------------------------|----------------|--------------|---------|-------------|---------|
| <i>Training participants’ own averages</i> | <i>human reference</i> | —              | 93.3         | 25      | 60          | 195     |
| Serial self-terminating (FIT)              | trial-level            | 9              | 80.5         | 14      | 71          | 157     |
| CGS, shared timing                         | trial-level            | 12             | 82.9         | 29      | 68          | 152     |
| CGS, per task                              | trial-level            | 24             | 98.8         | 24      | 84          | 189     |
| <b>gs-vision, refit</b>                    | <b>ACT-R run</b>       | 13             | <b>100.1</b> | 25      | 90          | 186     |
| Fixation-based (H&O)                       | trial-level            | 12             | 105.6        | 76      | 78          | 163     |
| GS6 engine, rates fitted                   | trial-level            | 12             | 114.3        | 22      | 100         | 222     |
| gs6-vision, rates fitted                   | ACT-R run              | 16             | 122.5        | 42      | 98          | 227     |
| GS6 engine, as posted                      | trial-level            | 6              | 123.3        | 121     | 78          | 171     |
| <b>gs-vision, frozen fit</b>               | <b>ACT-R run</b>       | 6              | <b>127.7</b> | 99      | 162         | 122     |

| Model                             | ACT-R status         | Fit parameters | Overall | Feature | Conjunction | Spatial |
|-----------------------------------|----------------------|----------------|---------|---------|-------------|---------|
| Parallel race                     | trial-level          | 14             | 129.4   | 72      | 131         | 185     |
| Stock ACT-R vision, timing fitted | timing mirror        | 6              | 137.3   | 29      | 171         | 211     |
| PAAV, fitted                      | display-level mirror | 9              | 275.4   | 49      | 109         | 668     |
| gs6-vision, engine as posted      | ACT-R run            | 8              | 312.9   | 92      | 304         | 543     |
| gs-vision, module defaults        | ACT-R run            | 0              | 365.4   | 103     | 265         | 729     |
| PAAV, posted values               | display-level mirror | 0              | 382.9   | 71      | 497         | 582     |
| Stock ACT-R vision, 4 finsts      | timing mirror        | 0              | 501.1   | 116     | 546         | 841     |
| Stock ACT-R vision, 20 finsts     | timing mirror        | 0              | 540.1   | 116     | 574         | 930     |
| gs6-vision, Wolfe's values        | ACT-R run            | 0              | 795.1   | 157     | 564         | 1,664   |

*Note.* Lower RMSE indicates a better fit. “ACT-R run” identifies results produced by an implemented ACT-R module. Mirror rows reproduce the relevant module mechanisms without running the original module. All post-freeze refits and baseline comparisons are exploratory; the refit row is the trigger candidate, while validation selection favored the frozen shared fit. The human-reference row scores the training participants’ cell averages against the test participants and is not a fitted model.

**Table 8***Slope and error measures for all models on the held-out test participants*

| Model                                      | Slopes within<br>5 ms/item | Largest miss<br>error (points) | Miss %     | False-<br>alarm % |
|--------------------------------------------|----------------------------|--------------------------------|------------|-------------------|
| Serial self-terminating (FIT)              | 3/6                        | 6.7                            | 3.2        | 0.85              |
| CGS, shared timing                         | 2/6                        | 6.1                            | 4.6        | 1.15              |
| <i>Training participants' own averages</i> | <i>5/6</i>                 | <i>2.0</i>                     | <i>3.5</i> | <i>1.18</i>       |
| CGS, per task                              | 2/6                        | 2.7                            | 3.6        | 1.87              |
| <b>gs-vision, refit</b>                    | <b>4/6</b>                 | <b>11.3</b>                    | <b>7.7</b> | <b>0.29</b>       |
| Fixation-based (H&O)                       | 3/6                        | 5.9                            | 5.0        | 0.01              |
| GS6 engine, rates fitted                   | 1/6                        | 4.2                            | 3.3        | 1.27              |
| gs6-vision, rates fitted                   | 3/6                        | 18.4                           | 11.1       | 1.30              |
| GS6 engine, as posted                      | 2/6                        | 12.0                           | 7.6        | 1.40              |
| <b>gs-vision, frozen fit</b>               | <b>2/6</b>                 | <b>10.9</b>                    | <b>7.2</b> | <b>0.00</b>       |
| Parallel race                              | 3/6                        | 8.6                            | 1.5        | 1.18              |
| Stock ACT-R vision, timing fitted          | 2/6                        | 10.0                           | 0.0        | 0.00              |
| PAAV, fitted                               | 3/6                        | 6.4                            | 2.3        | 0.00              |
| gs6-vision, engine as posted               | 4/6                        | 14.5                           | 8.0        | 1.45              |
| gs-vision, module defaults                 | 3/6                        | 12.4                           | 7.1        | 0.00              |
| PAAV, posted values                        | 2/6                        | 9.0                            | 8.4        | 0.00              |
| Stock ACT-R vision, 4 finsts               | 2/6                        | 18.4                           | 7.3        | 0.00              |
| Stock ACT-R vision, 20 finsts              | 2/6                        | 10.0                           | 0.0        | 0.00              |
| gs6-vision, Wolfe's values                 | 3/6                        | 10.8                           | 9.1        | 1.48              |

*Note.* Miss and false-alarm percentages are averages across the corresponding twelve task-by-set-size cells, with equal participant weighting within human cells. Human test means are 4.1% misses and 2.3% false alarms. The largest miss error is the largest absolute model-human difference across present-trial cells. The slope measure counts the six task-by-presence slopes that fall within 5 ms per item of the human slope.

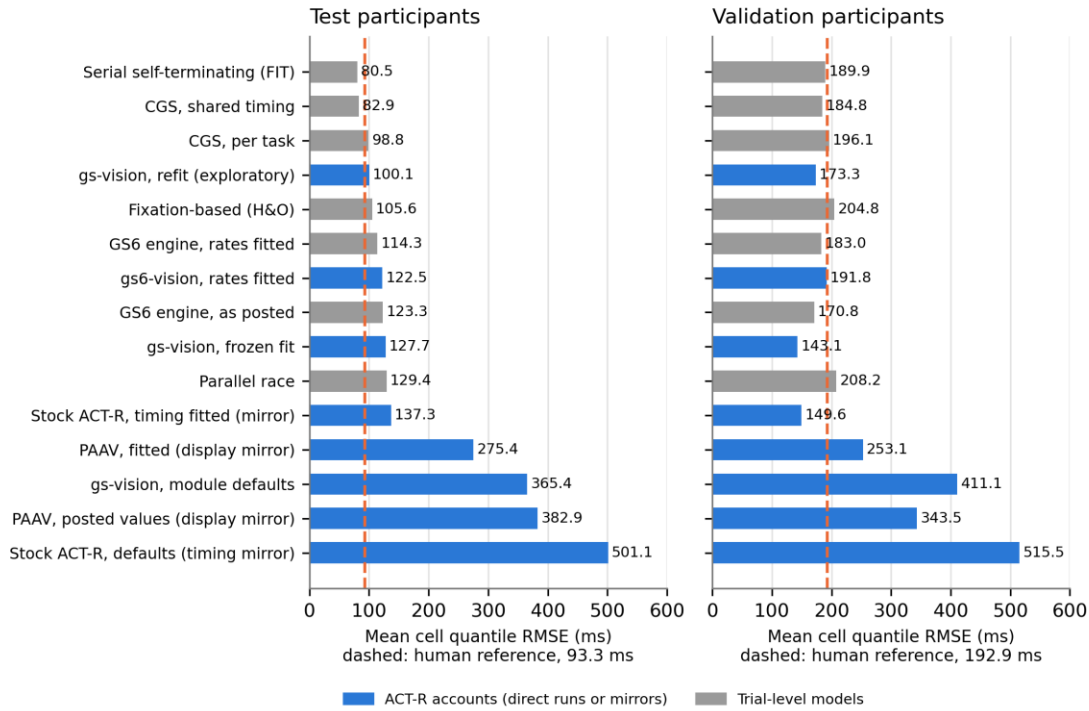


Figure 3. Mean cell quantile RMSE for 15 selected configurations on test participants (left) and validation participants (right). Blue bars denote ACT-R vision-module accounts, including direct gs-vision and gs6-vision runs, the stock timing mirror, and the PAAV display-level mirror. Gray bars denote trial-level models without displays. Dashed lines score the training participants' cell summaries against each evaluation split (93.3 ms for test and 192.9 ms for validation). These are descriptive cross-participant references, not uncertainty bounds or lower limits on model error.

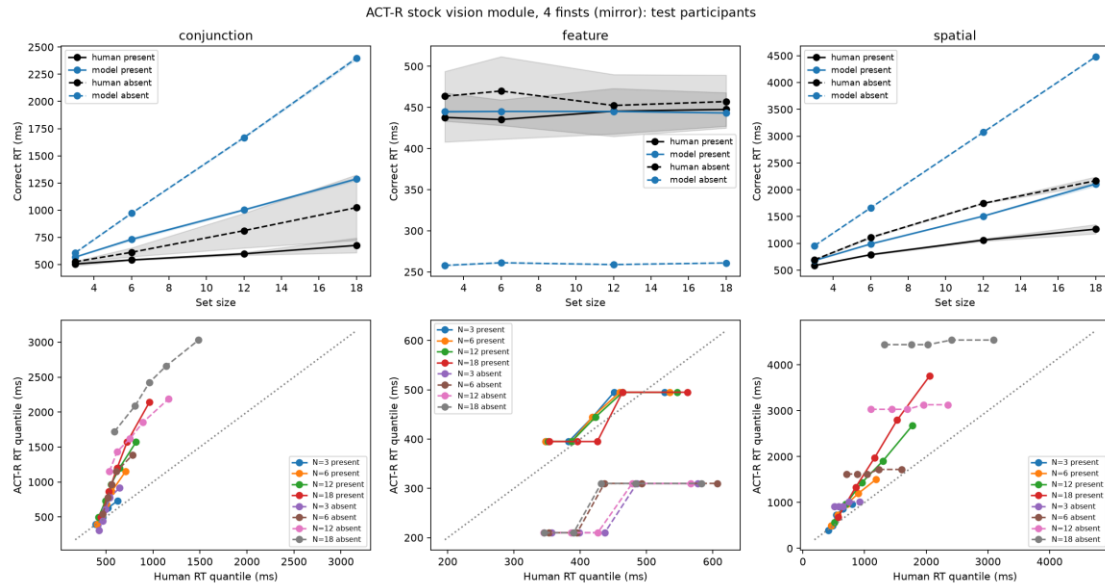


Figure 4. The stock ACT-R vision module driven by a find, attend, test production loop (timing mirror, four finsts), against the test participants, in the format of Figure 2.

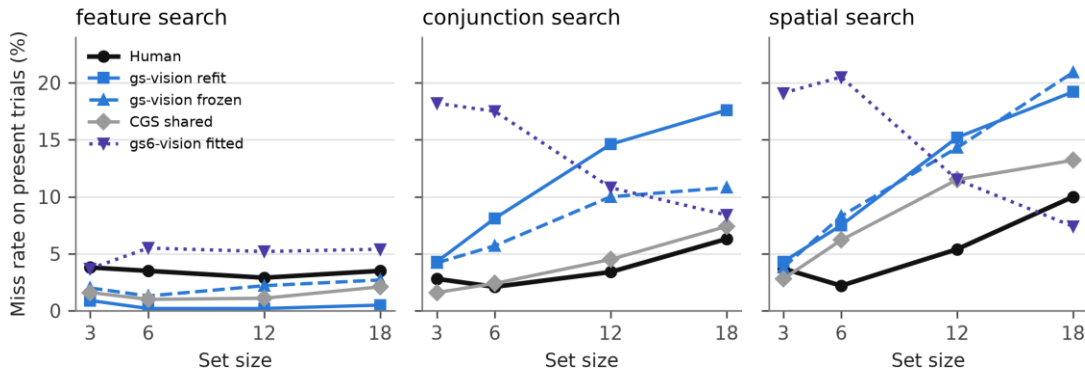


Figure 5. Miss rates on present trials by set size for the human test participants and four models.

Four conclusions follow from this exploratory comparison. First, gs-vision provides the lowest RT-quantile error among the ACT-R vision-module accounts tested here. On the held-out participants, the exploratory refit scored 100.1 ms, and the frozen fit scored 127.7 ms, compared with 122.5 ms for the fitted gs6-vision variant, 137.3 ms for the fitted stock-module timing mirror, 275.4 ms for fitted PAAV, 382.9 ms for PAAV at its posted values, and 501.1 to 540.1 ms for the stock module with its default timing. The exploratory trigger refit therefore has the

lowest numerical RT-quantile error among the ACT-R accounts in this comparison. The validation-selected frozen gs-vision fit does not outperform fitted gs6-vision on test RT quantiles. This numerical ranking does not generalize to every visual task or dependent measure.

Second, three trial-level configurations have lower numerical test quantile RMSE than the gs-vision refit: serial self-terminating search (80.5 ms), CGS with shared timing (82.9 ms), and CGS fitted separately by task (98.8 ms). The training participants' cell summaries, treated as predictions of test participants, score 93.3 ms. This reference measures cross-group variation; it is not an error floor or a significance threshold for differences between models. Rankings change on the validation split, and with only two test participants per task, the reported point estimates do not establish population-level superiority or equivalence. Published per-participant fits use a different protocol and should not be read as directly comparable error estimates.

Third, the models differ in what they explain. The two best trial-level configurations fit response-specific nondecision-time offsets and an exponential residual. The serial model also fits task-specific processing times. Meanwhile, CGS with shared timing shares its identification parameters across tasks and varies guidance and quitting by task. In contrast, gs-vision uses the response times produced by ACT-R's motor module and shares one identification process and one quit controller across tasks. It derives task differences from the display, acuity, and target template. gs-vision closely fits feature search and parts of the conjunction distribution, but its mean spatial RT is 110 to 225 ms slower than the test participants across the eight cells. Conjunction-absent distributions remain a misfit at set sizes 3 and 6: the model's 10th percentile is 24 ms earlier than the human value at set size 3 and 30 ms later at set size 6, while the means are 94 and 133 ms later, respectively.

Fourth, gs-vision's main limitation is its miss rate (Figure 5). Its largest cell-level miss-rate error is 11.3 percentage points, compared with 2.7 to 6.7 points for the best-fitting trial-level models. At set size 18 in conjunction and spatial search, gs-vision misses 18% to 19% of targets, whereas the human observers miss approximately 6% and 10%. The competitive quit rule can terminate a search while a target is being identified or while waiting for a saccade; the adaptive threshold rule instead waits for outstanding identifications to drain. CGS's task-specific quit weight follows the human increase in misses more closely. Revising how the competitive and adaptive quit mechanisms interact is therefore the most important next theoretical change.

### **Discussion and Conclusion**

This paper introduced gs-vision as an extension of ACT-R's stock vision module and provided a complete workflow for using it in cognitive models. The module integrates feature-based guidance, acuity, covert selection, capacity-limited identification, eye movements, memory, and search termination while retaining the existing ACT-R visual buffers and ordinary requests. In the Wolfe et al. (2010) benchmark, the exploratory gs-vision trigger refit produced the lowest numerical test RT-quantile error among the ACT-R vision-module accounts evaluated under common scoring criteria. The exploratory refit obtained a mean cell quantile RMSE of 100.1 ms, compared with 122.5 ms for gs6-vision, 137.3 ms for the fitted stock timing mirror, and 275.4 ms for fitted PAAV.

These results support consideration of gs-vision as the model's vision module when an ACT-R model requires a process-level account of visual search. At the same time, high miss rates, long saccades, weak prevalence effects, and the small number of held-out participants limit the scope of this conclusion. The public implementation and validation workflow make these

limitations testable and provide a foundation for improving visual perception in ACT-R models and for developing vision modules for other architectures.

### **What the module offers a modeler**

A model that uses gs-vision gains, for one buffer request, a search whose time course follows the display: efficient when a guiding feature separates the target, inefficient when only shape does, longer on absent trials than on present ones by a ratio that depends on the task, with positively skewed RT distributions whose spread grows with set size, miss rates that rise with set size, fixations that are logged with positions and durations, and learning across trials of a quitting criterion and feature priming. It gains these while keeping every stock request, every parameter, EMMA, the motor module, and the Environment, so that an existing model can adopt the module by loading it and replacing one production loop with one request. The module runs at about 3 ms of wall-clock time per trial, so experiments with tens of thousands of trials are practical.

For cognitive models of applied tasks, the relevant properties are that the search cost of an interface element is now a function of its features, its position relative to the eye and the features of its neighbors, rather than a constant. This guidance can be restricted or turned off to model an observer who does not know what the target looks like, and that a salience slot and a :gs-value-hook let a pixel-level saliency model (Itti et al., 1998) or a learned value map feed the priority map from outside, which is the path to natural-scene search benchmarks such as COCO-Search18 (Chen et al., 2021).

### **What it does not yet do**

We have reported the failures as measured, and we list them here because they are the roadmap. Miss rates rise too steeply with set size in conjunction and spatial search, despite the refit's fitted error goal of 0.12. Moreover, a per-task quit unit in the style of CGS is one

candidate revision. Compared with the unmatched foraging reference, saccade amplitudes are about twice as large and fixation durations are shorter, because covert selection exhausts the attentional field before the eye moves; a policy that lets the eye start earlier is implemented as :gs-saccade-trigger and was fixed at 6 degrees in the reported trigger refit, and matched scanpath comparisons await eye-tracking data with the timing and stimulus information a simulation needs. The frozen model produces no false alarms; the refit's decision-error parameter produces them at 0.3 percent against 2.3, and the Guided Search 6 variant's two-bound diffusion gets closer. The hybrid and the separate posted-GS6 replication both fail the intended joint prevalence pattern. The hybrid already uses a constant correct-rejection step, so removing prevalence scaling from that step is not a solution for this implementation. Changes to the feedback controller and task-specific error goals require direct evaluation. Priming of pop-out is not established. Finally, each training group contains only five or six participants, and each validation or test group contains two. Because the observed speed differences between splits limit the precision and generalizability of the comparison, leave-participants-out or hierarchical fitting is a logical next step.

### **Scope of the accuracy claim**

The results support the following claim: among the vision modules that were developed for ACT-R and evaluated on the Wolfe et al. (2010) benchmark, the exploratory trigger refit of gs-vision has the lowest numerical test RT-quantile error. Its exploratory refit reduces quantile RMSE by 27% relative to the fitted stock-module timing mirror, by 64% relative to fitted PAAV, and by 18% relative to the fitted gs6-vision variant. Validation selection leaned more favorably towards retaining the frozen shared fit, whose test error exceeds that of fitted gs6-vision. The comparison does not show that gs-vision is the best visual-search model in general. Three trial-

level configurations from two model families have lower numerical RT error, and some models reproduce misses more accurately. gs-vision offers a different combination of capabilities. First, it operates on a display, represents acuity and eye movements. Second, it uses ACT-R's measured response stage, and third provides a buffer interface for integration into a complete cognitive model. We therefore recommend it when those capabilities are needed, while treating its quitting and eye-movement predictions as targets for further development.

### **Declarations**

*Funding.* No funding was received for this work.

*Conflicts of interest.* The authors declare no competing interests.

*Ethics approval.* Not applicable; the study analyzed previously published, de-identified data.

*Consent to participate.* Not applicable.

*Consent for publication.* Not applicable.

*Availability of data and materials.* See the Open Practices Statement.

*Code availability.* See the Open Practices Statement.

*Authors' contributions.* AB: conceptualization, software, validation, writing the paper, editing.  
FER: reviewing, editing.

### **Open Practices Statement**

The gs-vision module, the gs6-vision variant, the Python mirror and harness, the trial-level baseline models, all fitting and evaluation scripts, the run records (parameter values, seeds, source hashes, commands and logs) for every table in this paper, and the generated evaluation artifacts are available at <https://github.com/ar-zadeh/gs-vision>. The human data are the Wolfe et al. (2010) archive ([https://search.bwh.harvard.edu/new/data\\_set\\_files.html](https://search.bwh.harvard.edu/new/data_set_files.html)) and the Wu and Wolfe (2022) fixation data (<https://osf.io/vzg28/>); the harness downloads them and verifies their

hashes. The following commands regenerate evaluation reports from the archived runs and execute the test suites. Refitting requires separate fit, selection, and final-simulation stages in `harness/repair.py`, `harness/refit.py`, `harness/gs6_fit.py`, and `harness/compare_models.py`; the archived run records specify their settings:

```
python harness/report.py --manifest data/model/repair_20260905/run_manifest.json --final-test
python harness/refit.py evaluate
python harness/gs6_fit.py evaluate
python harness/compare_models.py evaluate
python harness/compare_models.py report
sbcl --non-interactive --load tests/test_module_events.lisp
python -m pytest reference/test_reference.py tests/ -q
```

None of the reported studies were preregistered. The train/validation/test split, the objective, and the three summary targets were fixed and recorded in the repository before the first fit.

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### Appendix: The complete benchmark model

The file `models/search-model.lisp` defines the model used for every ACT-R result in this paper. One production per task issues the search (the three templates have different slots, and a request slot with a nil value is not the same as an absent slot), two productions respond, and one production reports the outcome. The two functions at the end are registered as remote commands for the harness.

```
;; Author: AmirBagherzadeh, 9/5/2026
(clear-all)
(define-model gs-search-model
  (sgp :v nil :trace-detail medium :er t
    :emma t :gs-enabled t
    :visual-num-finsts 4 :visual-finst-span 3.0)

  (chunk-type trial task state target-color target-orient target-shape outcome)
  (dolist (c '(start searching responded waiting done
    feature conjunction spatial bar digit two five))
    (unless (chunk-p-fct c) (define-chunks-fct `((,c name ,c)))))
  (define-chunks (goal isa trial task feature state waiting))
  (goal-focus goal)

  (p search-feature
    =goal> isa trial task feature state start target-color =c
    ?visual> state free
    ==>
    +visual> isa gs-search color =c
    =goal> state searching)

  (p search-conjunction
    =goal> isa trial task conjunction state start
    target-color =c target-orient =o
    ?visual> state free
    ==>
    +visual> isa gs-search color =c orient =o
    =goal> state searching)

  (p search-spatial
    =goal> isa trial task spatial state start target-shape =s
```

```

?visual> state free
==>
+visual> isa gs-search shape =s
=goal> state searching)

(p respond-present
=goal> isa trial state searching
=visual>
?manual> state free
==>
+manual> cmd press-key key "j"
-visual>
=goal> state responded)

(p respond-absent
=goal> isa trial state searching
?visual> state error
?manual> state free
==>
+manual> cmd press-key key "f"
=goal> state responded)

(p report-outcome
=goal> isa trial state responded outcome =o
?visual> state free
==>
+visual> isa gs-feedback outcome =o
=goal> state waiting outcome nil))

(defun gs-trial-setup (task color orient shape)
  (let ((task (string->name task))
        (color (and color (string->name color)))
        (orient (and orient (string->name orient)))
        (shape (and shape (string->name shape))))
    (dolist (c (remove nil (list task color orient shape)))
      (unless (chunk-p-fct c) (define-chunks-fct `((,c name ,c)))))
    (mod-focus-fct (list 'task task 'state 'start 'outcome nil
                        'target-color color 'target-orient orient
                        'target-shape shape))
      t))

(defun gs-trial-feedback (outcome)
  (let ((o (string->name outcome)))
    (if (member o '(hit miss fa tn))
        (progn (mod-focus-fct (list 'state 'responded 'outcome o)) t)
        (print-warning "gs-trial-feedback expects hit, miss, fa or tn, not ~s." outcome))))

(dolist (c '(("gs-trial-setup" gs-trial-setup "Arm the model for one trial.")
              ("gs-trial-feedback" gs-trial-feedback "Report a trial outcome.")))
  (unless (check-act-r-command (first c))
    (add-act-r-command (first c) (second c) (third c))))

```